

PRAHARI — AI-Based Predictive Personnel Stress and Welfare Monitoring System

Smart India Hackathon 2026 · Problem Statement 26186 Master Concept Document — v1.0

Praharī (प्रहरी) — "sentinel / one who keeps watch." The name is deliberate: the system watches over personnel, not on them.

0. How to read this document

This is the single source of truth for the project. Every team member should read it end to end at least once, then use it as a reference.

The document is organised so that:

- **Sections 1–4** explain *what we are building and why*. Non-technical. Read first.
- **Sections 5–8** explain *how the system is put together*. Architecture, data, modules.
- **Sections 9–11** are the *deep technical spine* — ML, privacy, design ethics.
- **Sections 12–16** are *practical* — stack, APIs, demo script, repo layout.
- **Section 17** is the *build plan*.

Every module carries a **build tier**:

Tier	Meaning
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P0	Must exist and be demoable. Without it the product doesn't make sense.
P1	Built after the P0 spine is stable. Adds depth.
P2	Designed and documented, not built . We defend these as the production roadmap. This is a legitimate answer at SIH provided we can explain the design in detail.

If you only remember one thing from this document, remember **Section 2.2 — the five non-negotiable principles**. Every design argument we have will be settled by them.

1. The problem in plain language

1.1 What is actually happening in the real world

Personnel in CAPFs (CRPF, BSF, CISF, ITBP, SSB, Assam Rifles), the Armed Forces, and state police work under conditions that are structurally stressful:

- Long deployments far from home, often in conflict or high-alert zones
- Irregular and unpredictable duty hours, frequent night duty, denied rest days
- Extended family separation — many personnel go 10–18 months without home leave
- Exposure to traumatic incidents (encounters, casualties, disaster response)
- Frequent transfers that break social support and disrupt children's schooling
- A culture where admitting psychological difficulty is perceived as weakness

The consequence is measurable and serious: elevated rates of stress-related illness, burnout, attrition, fratricide, and suicide within uniformed

services. These are not abstract statistics — they are the reason this problem statement exists.

1.2 How welfare is handled today

Almost entirely **reactively** and **manually**:

- A commanding officer or subedar notices someone "not behaving normally"
- Or the person self-reports to a unit welfare officer
- Or — most commonly — nothing is noticed until an incident occurs

Both existing pathways depend on human observation of an individual by a superior who may be supervising 100+ personnel, is not clinically trained, and whose attention is consumed by operational duties. Self-reporting is suppressed by stigma and by fear of career consequences.

The gap is not the intervention. Units already have welfare officers, counsellors, leave policy, and buddy systems. The gap is *knowing who needs them, early enough to matter.*

1.3 The core insight of this project

Stress leaves a trail in administrative records long before it becomes a crisis.

A person under sustained strain shows patterns in the data an organisation *already collects for other reasons*:

- Their leave requests get denied, or they stop applying at all
- Their consecutive duty days climb, their night-duty share rises
- Their short sick-leave frequency increases
- Their time-since-home-visit stretches past a year
- They get transferred again before settling
- Their unit experiences a critical incident and nothing changes in the roster afterwards

None of these is meaningful alone. **Together, and measured as a**

deviation from that individual's own normal, they form a signal.

This is the heart of our system. We are not building a lie detector or a mood scanner. We are building a system that reads administrative exhaust and says: *"This person's pattern has changed in a way that historically precedes welfare problems. Someone should talk to them."*

1.4 What the system is NOT

State this clearly and often, in the document, in the app, and in the presentation:

- ✘ It does **not** diagnose mental illness. It has no clinical authority.
 - ✘ It does **not** take any action automatically. Every action is taken by a human officer.
 - ✘ It does **not** feed disciplinary, promotion, or posting decisions. This is blocked in code.
 - ✘ It does **not** surveil communications, messages, calls, location, or social media.
 - ✘ It does **not** display a numeric "stress score" attached to a person's name.
 - ✘ It does **not** route distress reports up the chain of command by default.
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2. Product thesis and design principles

2.1 One-line thesis

Prahari converts routine HR and duty data into early, explainable, privacy-preserving welfare signals — producing a short, prioritised list of personnel who would benefit from a conversation, with the reason attached.

The output of the entire system, for a welfare officer on a Monday morning, is roughly this:

WELFARE CHECK-IN QUEUE — 3rd Bn, A Coy — Week 37

1. Personnel ID CRP-A-0417 Priority: HIGH
 Why: 14 consecutive duty days · night-duty share up 2.
 personal baseline (6 wks) · no home leave in 419
 2 leave applications declined
 Suggested: Prioritise leave sanction · duty rotation
 Confidence: High (rich data) · Flagged 4 days ago
2. Personnel ID CRP-A-0092 Priority: MEDIUM
 ...

Not a score. Not a diagnosis. **A name, a reason, and a suggested action.**

2.2 The five non-negotiable principles

These settle every design argument. If a feature violates one of these, we do not build it, no matter how impressive it looks.

P1 · Welfare, never discipline. The system exists to help people. Welfare data is technically firewalled from any disciplinary, promotion, ACR, or posting workflow. Disciplinary records are **excluded as model inputs** — if we fed punishment records into a stress model we would have built a punitive system by accident. We will demonstrate this block live.

P2 · Bands, never scores. No user ever sees "Stress: 78/100" next to a human being. Output is a band — *Stable / Monitor / Check-in recommended / Priority check-in* — plus the drivers behind it. Numeric scores exist internally for model evaluation only and are never rendered in a personnel-facing or command-facing view. Numbers invite ranking; ranking invites stigma.

P3 · Aggregate first, individual on justification. The dashboard opens on unit-level trends. To open an individual record, an officer must have

the role *and* type a reason, which is logged immutably. Aggregate views enforce a minimum cohort size ($k \geq 10$) so "aggregate" never leaks individuals.

P4 · The human decides, always. The system recommends. A person acts. No automated leave grant, no automated referral, no automated notification to a commander. Human-in-the-loop is mandatory and enforced at the API level — recommendations have a status of `pending_officer_decision` until a human resolves them.

P5 · Consent is granular, revocable, and visibly consequential. Personnel consent separately to HR analytics, self-reports, and biometric data. Revoking any of them visibly degrades the model's output and confidence for that person rather than silently retaining the data. There is a screen in the app called **"What my officer can see"** that is literally accurate.

2.3 Why these principles are the actual product

Any competent team can train a classifier. The difficulty in this problem statement — and the criteria the PS explicitly names — are:

"Preventing stigmatization of personnel identified as potentially at risk"
"Building trust among personnel regarding system usage and data protection"
"Ensuring ethical and transparent AI decision-making"

A system personnel refuse to use has zero accuracy in practice. **Trust is a functional requirement, not a soft one.** Our differentiator is that we treat it as engineering.

3. Users and what each one sees

Six roles. The *entire* privacy architecture is expressed in the differences

between these columns.

	Personnel (self)	Welfare Officer	Counsellor / Medical Officer	Unit Commander	Auditor
Own risk band	✓	—	—	—	—
Own raw self-reports	✓	✗	✓ (consent-gated)	✗	✗
Others' risk band	✗	✓ (own unit)	✓ (assigned cases)	✗	✗
Others' drivers/ reasons	✗	✓	✓	✗	✗
Others' raw self-report text	✗	✗	✓ (consent-gated)	✗	✗
Clinical instrument scores	✓ (own)	✗	✓	✗	✗
Unit aggregate trends	✗	✓	✗	✓	✗
"Welfare action in progress" flag	—	✓	✓	✓ (no reason shown)	✗
Access logs	✓ (who viewed me)	✗	✗	✗	✓

Log content / personal data	—	—	—	—	✗ (metadata only)
Infrastructure & model config	✗	✗	✗	✗	✗

3.1 Role narratives

Personnel (Constable / Jawan / Officer). Opens the app on a shared or personal phone, in Hindi. Takes a 25-second check-in. Sees their own trend, sees exactly what their officer can see about them, can message a counsellor confidentially, can revoke any consent instantly, can export or delete their data.

Welfare Officer. The primary operator. Opens the dashboard, sees the unit heatmap, works down a prioritised check-in queue of 3–8 people, reads the drivers, records what action was taken and the outcome. Cannot read anyone's private self-report text — only derived signals.

Counsellor / Medical Officer. Receives referrals and acute-distress routing. Sees clinical detail where the person consented. Sits *outside* the chain of command by design — this is the confidential channel that makes self-reporting safe.

Unit Commander (Coy Cdr / CO). Sees the health of the *unit*, not of individuals. Sees roster imbalance, aggregate trend, and how many welfare actions are open. Sees "welfare action in progress" against a slot but **never the reason**. Their lever is the duty roster and leave sanction, not the individual's psychology.

Auditor (Vigilance / DPO). Sees the immutable access log: who opened what, when, and the stated reason. Cannot see the content of any record. This role is what makes P3 real rather than decorative.

System Admin. Manages roles, model versions, thresholds, retention jobs. Sees zero personnel data — data is encrypted at field level and the admin holds no decryption authority.

4. Glossary

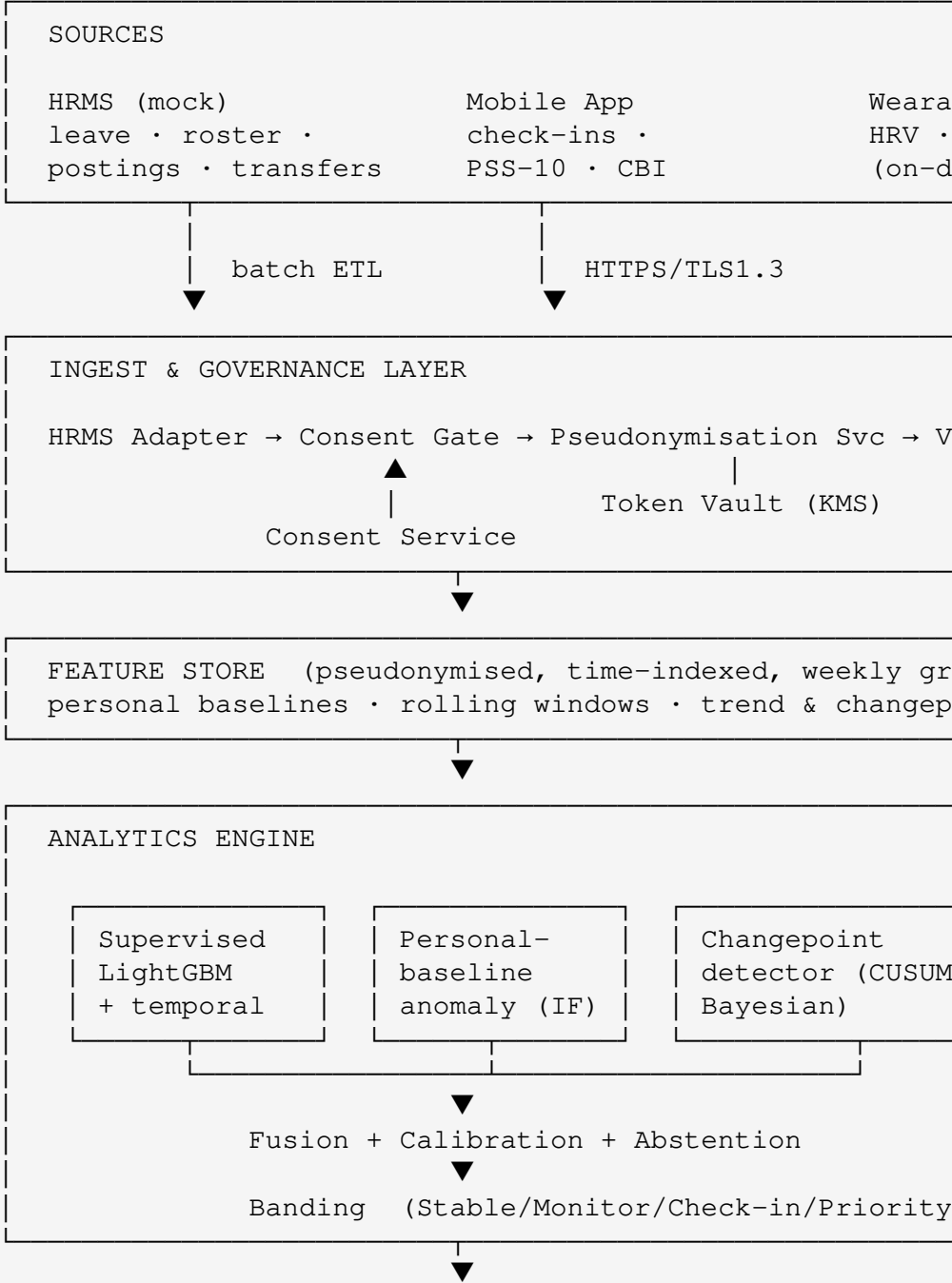
Read this once so nobody is lost in a discussion.

Term	Meaning
CAPF	Central Armed Police Forces — CRPF, BSF, CISF, ITBP, SSB, Assam Rifles
HRMS	Human Resource Management System — the existing system holding leave, posting, duty records
Baseline (personal)	An individual's own historical normal for a metric. We compare people to themselves, not to each other.
Changepoint	A statistically detected moment where a time series shifts regime. Our core "something changed" primitive.
CUSUM	Cumulative Sum control chart — a lightweight changepoint detector
Risk band	The banded output: Stable / Monitor / Check-in / Priority check-in
Driver	A plain-language reason contributing to a band, e.g. "no home leave in 419 days"
PSS-10	Perceived Stress Scale, 10-item. Validated, free, 10 questions. Our primary stress self-report.
CBI	Copenhagen Burnout Inventory. Free to use. Our burnout instrument. (We avoid MBI — it is licensed.)
WHO-5	WHO-5 Well-Being Index. 5 items, free, excellent screening properties.

PHQ-9 / GAD-7	Clinical depression / anxiety screeners. Optional, counsellor-mediated, never on a dashboard.
HRV	Heart Rate Variability — physiological stress correlate from wearables
Pseudonymisation	Replacing service numbers with tokens; the mapping lives in a separate vault
k-anonymity	Guarantee that any displayed group contains $\geq k$ individuals, so no one is identifiable
Differential privacy	Adding calibrated noise to published statistics so individuals cannot be inferred
Federated learning	Training a model where data stays on the device; only model updates travel
SHAP	SHapley Additive exPlanations — attributes a prediction to its input features
DPDP Act 2023	India's Digital Personal Data Protection Act — the law we must comply with
Data Fiduciary	Under DPDP, the entity deciding purpose/ means of processing. Here: the Force.
Break-glass	Emergency override access, permitted but heavily logged and reviewed after
PR-AUC	Precision–Recall Area Under Curve. The right metric for rare events (not ROC-AUC).
Lead time	Days between our flag and the welfare event. Our headline metric.

5. System architecture

5.1 High-level flow



EXPLAIN → RECOMMEND → ROUTE

SHAP → plain-language drivers



Recommendation Engine (rules + resource-aware prioritization)



Alert Router (tiered; acute bypasses chain of command)



Welfare Dashboard
(aggregate-first,
k-anon enforced)



Counsellor Console
(clinical, consent-
gated, off-chain)

Personal Health
(own
transaction)

▲ every access ▲ every access ▲ every access
└──────────┘ AUDIT LOG (hash-chained, append-only)

5.2 Why this shape

Three architectural decisions carry most of the weight:

Consent gate sits *before* pseudonymisation and storage. Data that has no consent never enters the feature store at all. This is the difference between "we filter it out later" and "we never had it."

Pseudonymisation before the feature store. Everything downstream of ingest — the entire analytics engine, the models, the explainability layer — operates on tokens, never on service numbers. Re-identification is a separate, logged service call available only to the dashboard render layer for authorised roles.

The audit log is a peer of the application, not a feature of it. Every read of an individual record emits an audit event. It is hash-chained so tampering is detectable.

5.3 Deployment profile

Designed for **on-premise / air-gapped** deployment inside force

networks. No third-party cloud AI APIs in the data path. All models run locally. Data residency: India, on force infrastructure. For the SIH demo we run in Docker Compose locally, which conveniently *is* the air-gapped story.

6. Data model

6.1 Core entities

We keep the schema deliberately close to what a real HRMS would expose, so the integration story is credible.

personnel	# identity – lives in th
personnel_id (PK)	# internal UUID
service_number	# ENCRYPTED, field-level
name	# ENCRYPTED, field-level
rank	
unit_id (FK)	
date_of_joining	
home_district	
current_station	
marital_status	
dependents_count	
personnel_token	# the pseudonymisation m
token (PK)	# e.g. "PRH-7f3a91c2"
personnel_id	# ENCRYPTED with vault k
created_at	
unit	
unit_id (PK)	
unit_name	# "3 Bn CRPF, A Coy"
parent_unit_id	
unit_type	# operational / static /
posting_intensity	# high / medium / low

```

leave_record
  leave_id (PK)
  token (FK)                                # pseudonymised
  leave_type                                # EL / CL / sick / emerg
  applied_on
  from_date, to_date
  status                                    # sanctioned / declined
  is_home_visit (bool)
  cancelled_by                              # self / admin ← import

duty_assignment
  duty_id (PK)
  token (FK)
  duty_date
  shift                                    # day / night / split
  duty_type                                # patrol / static guard
  hours
  is_rest_day_violation (bool)
  roster_changed_within_24h (bool)

deployment
  deployment_id (PK)
  token (FK)
  location_category                          # LWE / border / IS duty
  intensity                                  # high / medium / low
  from_date, to_date
  distance_from_home_km

transfer_record
  transfer_id (PK)
  token (FK)
  from_unit, to_unit
  order_date, effective_date
  is_requested (bool)                        # self-requested vs impo
  family_moved (bool)

training_record
  token, course_name, from_date, to_date, is_residential

```

critical_incident	# unit-level, NOT indivi
incident_id (PK)	
unit_id (FK)	
incident_date	
incident_type	# casualty / encounter /
severity	

— consented, personnel-contributed —

checkin	# the 25-second daily/we
checkin_id (PK)	
token (FK)	
submitted_at	
mood_1_5, sleep_quality_1_5, energy_1_5, workload_feel_	
free_text	# ENCRYPTED, counsellor-

assessment	# validated instruments
assessment_id (PK)	
token (FK)	
instrument	# PSS10 / CBI / WHO5 / P
administered_at	
raw_score	
subscale_json	

clinical_assessment	# PHQ-9 / GAD-7 / PCL-5
...	# counsellor access only

wellness_signal	# derived from wearable,
token, date, hrv_rmssd, resting_hr, sleep_minutes, slee	

— governance —

consent	
consent_id (PK)	
token (FK)	
scope	# HR_ANALYTICS / SELF_RE
granted (bool)	
granted_at, revoked_at	
policy_version	# consent is versioned
expires_at	# consent is time-bound

```

audit_event                                # append-only, hash-chain
  event_id (PK)
  actor_user_id
  actor_role
  action                                  # VIEW_INDIVIDUAL / EXPOSE
  target_token
  stated_reason                          # MANDATORY for individuals
  timestamp
  prev_hash, this_hash

risk_assessment                           # model output
  assessment_id (PK)
  token (FK)
  as_of_week
  band                                   # STABLE / MONITOR / CHANGING
  internal_score                         # NEVER rendered to a user
  confidence                            # HIGH / MEDIUM / LOW / UNKNOWN
  drivers_json                           # [{feature, direction, value}]
  model_version
  expires_at                             # flags EXPIRE — no permanent flags

intervention
  intervention_id (PK)
  token (FK)
  recommended_action
  status                                # pending_officer_decision
  decided_by, decided_at
  action_taken_notes
  outcome                               # improved / unchanged / worse
  follow_up_date

```

6.2 Two schema decisions worth defending

Flags expire. `risk_assessment.expires_at` exists so no one carries a permanent "at-risk" label. A flag from 8 months ago is not a property of the person.

internal_score is column-level restricted. It exists for model evaluation and is excluded from every API serializer that feeds a UI. This is enforced in code, not by convention.

7. The synthetic data generator

Tier: P0. This is the foundation of the entire project.

7.1 Why we must build this

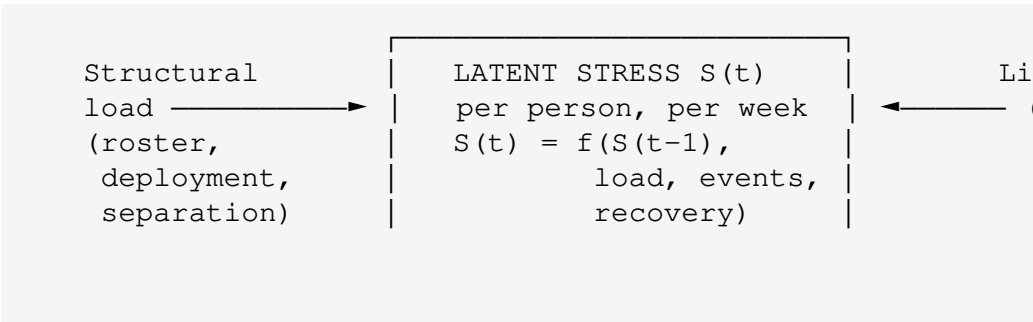
There is no public dataset that pairs uniformed-forces HR administrative records with wellness outcomes. It does not exist, and it will not exist for us. So we generate one.

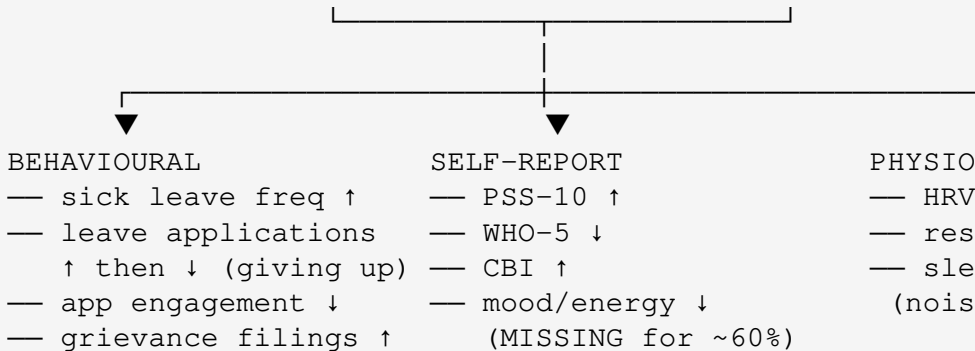
This is not a shortcut or a weakness — **it is a deliverable in its own right**, and how honestly we do it determines our credibility with judges. A generator that just sprays random numbers produces a model that learns nothing real. A generator with **defensible causal structure** produces a dataset where stress genuinely *causes* the administrative patterns we then try to detect — which means our detector is being tested on a meaningful task.

We document the generative assumptions openly and cite them.

7.2 The causal design

We simulate a latent variable and let observables flow *from* it. This direction matters enormously.





Key properties we encode:

1. **Stress is autoregressive with recovery.** $S(t) = \alpha \cdot S(t-1) + \beta \cdot \text{load}(t) - \gamma \cdot \text{recovery}(t) + \epsilon$. Home leave, rest days, and a stable posting drive *recovery*. This is why granting leave makes the signal fall — and why the demo works.
2. **Load is structural, not random.** Night-duty ratio, consecutive duty days, rest-day violations, deployment intensity, days-since-home-visit and family separation all feed *load(t)* with different weights.
3. **Heterogeneous baselines.** Each simulated person gets a resilience parameter and a *personal normal* for every metric. A person who always works nights has a high baseline night ratio and is not thereby stressed. This forces the model to learn deviation, not level.
4. **Non-monotonic help-seeking.** As stress rises, leave applications first increase, then *decrease* (learned helplessness / giving up). This is the kind of realistic non-linearity that makes the modelling problem interesting rather than trivial.
5. **Unit-level shocks.** Critical incidents raise *load* for an entire unit simultaneously, producing the cluster pattern our unit-level demo scenario needs.
6. **Realistic missingness.** ~60% of personnel never self-report. Wearable data exists for ~15% and is intermittent. Some HR fields

are missing or late. The model must work for the silent majority — which is exactly why the unsupervised track exists.

7. **Ground-truth welfare events.** We emit a labelled `welfare_event` when a person's $S(t)$ crosses a high threshold and stays there. **This label is used only for evaluation, never as a model feature.** It is what lets us measure lead time.

7.3 Output

- ~2,000 simulated personnel across ~8 units, 104 weeks of history
- All tables from Section 6 populated
- A `ground_truth.csv` with the latent stress trajectory and welfare events, kept out of the training path
- A config file exposing every parameter, so we can generate variants (e.g. a high-stress border battalion vs a static CJSF unit)

7.4 Tech

Python: `numpy`, `pandas`, `scipy`, `Faker` for names/service numbers, `pyyaml` for config. Written as `packages/datagen/` with a CLI:
`python -m datagen --config configs/crpf_bn.yaml --out data/.`

7.5 Validation of the generator itself

Before trusting it, we check: do the marginal distributions look plausible (leave days/year, night-duty share, transfer frequency)? Do the correlations between latent stress and observables match published occupational-health literature in sign and rough magnitude? Do simple baselines get non-trivial but non-perfect performance? If a logistic regression scores 0.99 PR-AUC, our generator is too easy and we tune it down.

8. Module specifications

Twelve modules. Each: what it does, why, how it's built, tech, tier.

M1 · Synthetic Data Generator & Mock HRMS — P0

What. Generates the synthetic population (Section 7) and serves it through a mock HRMS REST API that mimics a real personnel system.

Why. Two purposes: it gives us data, and the mock API proves the integration story. In a real deployment we'd point the adapter at the force's actual HRMS; the adapter interface stays identical.

How.

- `datagen` package produces CSV/Parquet + seeds a Postgres schema
- A small FastAPI service (`mock-hrms`) exposes endpoints like `GET /api/v1/personnel/{id}/leave?from=&to=`, deliberately in a slightly *awkward* shape (paginated, `snake_case` mixed with `camelCase`, dates as strings) so our adapter has real work to do

Output. Populated `hrms_raw` schema + a running mock service on `:8100`

M2 · HRMS Adapter, ETL Pipeline & Pseudonymisation — P0

What. Pulls from the source system, validates, applies the consent gate, replaces identifiers with tokens, and lands clean data.

Why. The PS explicitly requires "secure integration with HRMS and personnel management systems." This module is that requirement.

How.

1. **Adapter interface** — an abstract `HRMSAdapter` base class with `fetch_leave()`, `fetch_duty()`, `fetch_postings()`. `MockHRMSAdapter` implements it. A real deployment writes `CRPFHRMSAdapter` and changes one line of config. Present this as the adapter pattern — it's what makes the integration credible without access to a real system.
2. **Data contract** — a Pydantic schema per entity. Anything failing validation goes to a quarantine table with a reason, never silently dropped.
3. **Consent gate** — for each personnel record, query the Consent Service. No `HR_ANALYTICS` consent → the record is not ingested at all.
4. **Pseudonymisation** — look up or mint a token; write the mapping to the Token Vault (a separate schema, separate credentials, envelope-encrypted). All downstream tables carry only the token.
5. **Idempotent, incremental** — watermark-based, re-runnable.

Tech. Python, Pydantic v2, SQLAlchemy, Prefect (or plain APScheduler for the demo), `cryptography` for envelope encryption.

M3 · Feature Store & Feature Engineering — P0

What. Turns raw event tables into a weekly, per-person feature vector, including personal baselines and trend features.

Why. This is where domain insight actually lives. Model choice matters less than these features.

How. A weekly job materialises `features_weekly(token, week_start, ...)`.

Feature families — build all of these:

Leave signals

- `days_since_last_home_visit` ← single strongest expected predictor
- `leave_applications_declined_90d`,
`leave_cancelled_by_admin_90d`

- short_sick_leave_count_90d, sick_leave_trend_slope
- leave_taken_vs_entitled_ratio_ytd
- leave_application_rate_delta (captures the "gave up applying" pattern)

Duty roster signals

- consecutive_duty_days, max_consecutive_duty_days_30d
- night_duty_ratio_28d, night_duty_ratio_vs_personal_baseline
- rest_day_violations_30d
- roster_volatility (std-dev of shift type over 28d)
- last_minute_roster_changes_30d
- total_duty_hours_28d, duty_hours_trend_slope

Deployment signals

- days_in_current_high_intensity_posting
- consecutive_operational_deployments
- days_since_non_operational_posting
- weeks_since_unit_critical_incident, unit_incident_severity

Posting & family signals

- transfers_24m
- family_separated (bool), distance_from_home_km
- pending_transfer_request_age_days
- months_at_current_station

Administrative signals

- grievances_filed_180d, pending_applications_count
- welfare_fund_requests_180d

Self-report signals (when consented)

- checkin_completion_rate_28d — **engagement drop is itself a signal**
- mood_ma7, mood_slope_28d, sleep_quality_slope_28d

- `latest_pss10`, `pss10_delta`, `latest_who5`, `latest_cbi`
- `days_since_last_checkin`

Derived — the important ones

- For every continuous feature: `z_vs_personal_baseline` computed over that individual's trailing 26 weeks
- `changepoint_detected` and `weeks_since_changepoint` per key series (CUSUM)
- `composite_load_index` — a transparent weighted sum used as an interpretable fallback

Tech. Pandas/Polars for computation, Postgres tables as the store, `ruptures` or a hand-rolled CUSUM for changepoints. A simple `FeatureSpec` registry so features are declared once and reused by training and inference (prevents train/serve skew).

M4 · Risk Prediction Models — P0

What. Produces, per person per week, a calibrated risk estimate + confidence, fused from three tracks.

Why. Three tracks because the label situation is uneven — most people never self-report.

How.

Track A — Supervised (LightGBM). Trained where self-reports exist. Target: does this person's PSS-10 / CBI cross a concern threshold in the *next* 2–4 weeks? Tabular gradient boosting is the right call: strong on this data shape, fast, and natively SHAP-explainable. Grouped time-series CV split by person *and* time — never random split, or we leak.

Track B — Personal-baseline anomaly (unsupervised). Isolation Forest over the `z_vs_personal_baseline` feature block, plus a small sequence autoencoder (P1) over 12-week windows. Covers the ~60% who never self-report. Output is a deviation magnitude.

Track C — Changepoint detection. CUSUM/Bayesian changepoint on

key series. Not a model so much as a trigger; contributes strongly to *lead time*, since it fires on the shift rather than on the level.

Fusion. A transparent weighted combination with track-specific weights, then **isotonic calibration** so the number means something. Rationale: we can explain a weighted fusion to a commander; we cannot explain a stacked meta-learner.

Abstention. If a person has < 8 weeks of data or coverage below threshold, output `INSUFFICIENT_DATA` — never a low-confidence guess. This directly addresses the PS's false-positive concern.

Banding. Calibrated probability → band via thresholds tuned to a *capacity target*: "surface the top N per battalion per week where N matches counsellor availability." Thresholds are configurable per unit and stored in the admin console, not hard-coded.

Tech. scikit-learn, LightGBM, PyTorch (autoencoder, P1), `ruptures`, MLflow for experiment tracking and model versioning.

M5 · Explainability Layer — P0

What. Converts a model output into 2–4 plain-language drivers a non-technical welfare officer can act on.

Why. The PS names "ethical and transparent AI decision-making" as a key challenge. Also purely practical: an officer cannot act on "risk 0.71." They can act on "no home leave in 419 days."

How.

- TreeSHAP on the LightGBM model → per-feature contributions
- For the unsupervised track, feature-wise deviation magnitude
- A **driver template registry** maps feature name + direction + magnitude to a Hindi/English sentence:

```
days_since_last_home_visit:  
  direction: high  
  en: "No home leave in {value} days"
```

```
hi: "पिछले {value} दिनों से घर की छुट्टी नहीं"  
suggests: [GRANT_HOME_LEAVE]  
night_duty_ratio_vs_personal_baseline:  
direction: high  
en: "Night duty share is {value}× their usual over  
suggests: [DUTY_ROTATION, ROSTER_REBALANCE]
```

- Select top drivers by contribution, **filter to actionable ones first** — an officer can change the roster; they cannot change `months_since_joining`
 - Never render a force plot or a SHAP waterfall to an officer. Sentences only.
 - **Counterfactual hint (P1)**: "If home leave were sanctioned, this flag would likely clear."
-

M6 · Recommendation Engine — P0

What. Maps drivers to concrete welfare actions and produces a resource-aware, prioritised queue.

Why. A flag with no suggested action pushes the whole cognitive load onto the officer. Also, one counsellor per battalion is the real-world constraint — 200 simultaneous referrals is a useless output.

How.

- **Intervention catalogue** (configurable): grant home leave · duty rotation · temporary lighter posting · rest-day enforcement · buddy/peer pairing · counsellor referral · family liaison contact · financial counselling · unit debrief after critical incident · workload redistribution · medical review
- **Rules-driven mapping** (driver → action), stored as YAML so it is auditable and editable by the force without a code change. Deliberately *not* ML — recommendations must be traceable.
- **Prioritisation** ranks by band × driver actionability × time-since-flag × resource availability
- **Capacity awareness** — the engine knows how many counsellor slots and leave quotas exist and will not recommend beyond

them; it degrades to lower-cost interventions (peer pairing, rest-day enforcement) when clinical capacity is exhausted

- **Unit-level recommendations** — if > 30% of a sub-unit is amber, the engine emits a *unit* recommendation (roster rebalance, debrief) instead of N individual ones. This is one of our strongest demo moments.
 - **Outcome capture** — the officer records what was done and what happened. This is the only real label source a deployed system will ever generate, and it feeds retraining.
-

M7 · Alert Router — P0

What. Decides who gets told what, when, and through which channel.

Why. Routing *is* the privacy model in motion. Get this wrong and you have built a system that reports distress to a person's boss.

How — tiered routing:

Tier	Trigger	Goes to	Commander sees
Routine	Band = Monitor	Weekly digest to Welfare Officer	Aggregate only
Elevated	Band = Check-in	Task to Welfare Officer, 48h SLA	"Welfare action in progress"
Priority	Band = Priority check-in	Welfare Officer + Counsellor	"Welfare action in progress", no reason
Acute	Self-reported acute distress / crisis keyword	Counsellor directly, bypassing chain of command	Nothing

Alert fatigue controls — non-optional, because an ignored system is a failed system:

- Deduplication: one open alert per person at a time
- Suppression while an intervention is open
- Rate limiting per officer per week
- Escalation only on non-response, never on repeat detection

Crisis pathway is a separate hardened flow: immediate counsellor notification, in-app display of helpline numbers (Tele-MANAS 14416), a defined SOP, and human confirmation at every step. No automated escalation to command.

Channels. In-app + email + SMS fallback for low-connectivity postings.

M8 · Consent Service — P0

What. The system of record for who has consented to what, at which policy version, until when.

Why. Principle P5, and DPDP Act compliance. Also our best demo moment (Scenario 5).

How.

- **Scopes:** `HR_ANALYTICS`, `SELF_REPORT`, `BIOMETRIC`, `RESEARCH_AGGREGATE` — independently grantable and revocable
 - **Versioned:** consent is tied to a policy version; a policy change requires re-consent, it does not silently carry over
 - **Time-bound** with `expires_at` and a renewal prompt
 - **Revocation is real:** on revoke we (a) stop ingesting that scope, (b) delete the scoped raw data, (c) recompute the person's risk assessment without it, (d) mark confidence `REDUCED`, (e) log the whole thing
 - Every other service calls the Consent Service before touching data. It is a hard dependency, not a checkbox.
 - Exposes `GET /consent/{token}/effective` returning the currently active scopes
-

M9 · RBAC, Audit & Purpose Limitation — P0

What. Enforces the access matrix in Section 3, logs every access immutably, and technically blocks welfare data from reaching career-affecting systems.

Why. Principles P1 and P3. The PS names privacy as its first technical challenge.

How.

- **Roles** as in Section 3, with permissions expressed as `(role, resource, field-set)` so a Welfare Officer's API response literally does not contain the fields they may not see — filtered server-side at serialization, not hidden in the UI
 - **Reason-for-access:** `GET /personnel/{token}/detail` requires a non-empty `X-Access-Reason` header. Missing → HTTP 428. Present → audit event written before the data is returned.
 - **Hash-chained audit log:** `this_hash = SHA256(prev_hash + event_payload)`. A verification endpoint walks the chain and proves no tampering. Append-only enforced at the DB level (no UPDATE/DELETE grant on the table).
 - **Purpose limitation, enforced:** a `PurposeGuard` middleware tags every data export with its declared purpose and hard-rejects any request whose purpose is `PROMOTION`, `POSTING`, `DISCIPLINARY`, or `ACR`. We expose a deliberately-included endpoint that *attempts* this and gets blocked — so we can demo it.
 - **Break-glass:** an emergency-access role that works, but fires a notification to the auditor and requires post-hoc justification within 24h.
 - **Personnel see their own access log** — "who looked at my record and why." Very few systems do this, and it is worth a lot of trust.
-

M10 · Welfare Dashboard (Web) — P0

What. The primary operator interface for welfare officers, commanders, counsellors and auditors.

How — screens:

1. **Unit Overview (default landing)** — heatmap of sub-units × 12-week trend, band distribution, roster-imbalance panel. **k-anonymity enforced:** cells with cohort < 10 render as "insufficient cohort."
2. **Check-in Queue** — the prioritised list. Each card: pseudonymous ID (name revealed only on justified open), band chip, 2–4 drivers, confidence, suggested action, "Open" button.
3. **Individual View** — gated behind the reason-for-access modal. Shows band history, drivers over time, intervention history. **No numeric score anywhere.**
4. **Intervention Tracker** — kanban of open/accepted/completed actions with outcome capture and follow-up dates.
5. **Workload Distribution** — night-duty share, consecutive duty days and leave backlog across the unit. This is the commander's actual lever.
6. **Counsellor Console** — case list, consent-gated clinical detail, secure message thread with personnel.
7. **Audit View** — access log with filters and a "verify chain integrity" button.
8. **Admin Console (P1)** — role provisioning, model version, band thresholds, retention jobs, consent report.

Design rules. Calm, low-alarm visual language. Neutral vocabulary — "check-in recommended," never "at-risk." No red-alarm aesthetics for a human being. No leaderboards, no rankings, no sorting personnel by score. Bilingual EN/HI.

Tech. React 18 + TypeScript + Vite, TailwindCSS, TanStack Query, Recharts, shadcn/ui.

M11 · Personnel Mobile App — P0 (3 screens deep)

What. The personnel-facing app: consent, check-ins, transparency,

confidential help.

Why. Without this the system is pure surveillance. This app is what makes it voluntary and legitimate.

How — screens, in priority order:

1. **Onboarding & Granular Consent** — four independent toggles with plain-language explanations of what each enables and what happens if it's off
2. **Daily Check-in** — 4 sliders, 25 seconds, with a "skip today" that carries no penalty
3. **"What My Officer Can See"** — the trust screen. Renders literally the payload a welfare officer would receive about this user. Non-negotiable, and almost nobody builds it.
4. **My Trends** — the person's own data back to them, framed supportively
5. **Talk to Someone** — confidential counsellor channel, routed outside the chain of command, plus Tele-MANAS 14416
6. **Assessments** — PSS-10 / WHO-5 / CBI on a longer cycle, optional
7. **Self-help library** — psychoeducation, breathing/grounding, sleep hygiene, family-separation coping. Works offline.
8. **Privacy Centre** — revoke consent, view my access log, export my data, delete my data

Hard engineering constraints:

- **Offline-first.** Deployments happen where there is no network. Local SQLite queue, sync on reconnect, conflict resolution by timestamp. This is a real constraint, not a nice-to-have.
- **Multilingual.** Hindi + English minimum, architecture ready for regional languages, icon-supported for low literacy.
- **Low-end device support.** Android 8+, small APK.

Tech. React Native + Expo, TypeScript, `expo-sqlite`, `i18next`, `expo-secure-store` for tokens.

not code)

The PS names trust-building as a technical challenge. These are graded deliverables, and they cost days rather than weeks:

- **Model Card** — intended use, out-of-scope use, training data, metrics, fairness audit, limitations
 - **DPIA** (Data Protection Impact Assessment) — risks, mitigations, residual risk
 - **DPDP Act 2023 mapping table** — clause by clause against our implementation
 - **Personnel Privacy Notice** — plain Hindi and English, one page, readable by a constable
 - **Human-in-the-loop policy** — written statement that no automated action is taken on a person
 - **Appeals mechanism** — a documented route for a person to contest a flag, and a UI stub for it
 - **Escalation SOP** — what a welfare officer does at each tier, including the crisis pathway
 - **Threat model** — STRIDE table over the data flow
 - **Anti-stigmatisation design standard** — the vocabulary and UI rules in Section 11
-

9. The machine learning problem, honestly stated

9.1 The label problem — our central obstacle

We are asked to predict stress with no ground truth. Every team attempting this PS faces it. Most will pretend it isn't there. **We address it directly in the pitch, because doing so reads as maturity while hand-waving reads as naivety.**

Three honest answers:

- (a) Self-reports as proxy labels.** Where a person consents and completes PSS-10 / CBI / WHO-5, we have a validated, published, psychometrically-sound label. Not perfect — self-report has its own biases — but defensible. This trains Track A.
- (b) Unsupervised deviation for everyone else.** The ~60% who never self-report get personal-baseline anomaly detection, which needs no labels at all. This is not a compromise; for the silent majority it is the *only* correct approach.
- (c) Intervention outcomes as the real, long-run label source.** Once deployed, every welfare action recorded with an outcome is a genuine label. We design the outcome-capture loop from day one so that a real deployment starts generating training data immediately. We say this to judges explicitly: *"Version 1 ships with synthetic-validated models and a labelling loop; version 2 is trained on the force's own outcomes."*

9.2 Public datasets we validate components against

We cannot validate the HR-analytics track on public data — nothing like it exists. But we **can** validate the physiological and behavioural-sensing components on real data, which is far better than validating nothing:

Dataset	What it gives us	What we validate
WESAD	Chest + wrist physiology, labelled baseline/stress/amusement, 15 subjects	Our HRV/EDA stress-detection pipeline on real physiology
SWELL-KW	Knowledge-worker stress: physiology + computer-interaction logs + subjective ratings	Fusing behavioural logs with physiology — closest structural analogue to our HR+self-report fusion

Nurse Stress Prediction (Empatica E4)	Shift-working healthcare staff, real occupational stress, wearable + self-report	Closest <i>occupational</i> analogue: shift work, irregular hours, high-stakes environment
StudentLife	Passive behavioural sensing → mental-health outcomes over a term	Conceptually the closest published work to what we're doing: behaviour patterns predicting wellbeing over time
PMDData	Long-horizon lifelogging: sleep, activity, self-report	Long-window trend and baseline modelling

The honest framing for judges: physiological and behavioural components are validated on real public data; the HR-analytics component is validated on a causally-structured synthetic corpus whose assumptions are documented; field validation requires a pilot. That is exactly what a responsible team would say.

9.3 Evaluation — what we actually report

Accuracy is the wrong metric and we should never say it. Our evaluation suite:

Metric	Why
PR-AUC	Heavily imbalanced problem — ROC-AUC will flatter us and mislead
Precision / Recall at operating point	The officer's actual experience of the system
Lead time (days)	☆ Headline metric. How many days before a welfare event did we flag? A model that flags on the day of crisis is worthless.
Calibration: Brier score + reliability curve	If we say 70%, it must be right ~70% of the time

False alarms per 100 personnel per month	Determines whether officers keep using the system. This is the adoption metric.
Fairness audit	Equal-opportunity difference across rank, age band, region, unit type. A model that flags one region 3× more is broken.
Abstention rate	What fraction we correctly refuse to score
Coverage	What fraction of personnel we can meaningfully assess

We report a **baseline comparison** too: a transparent rules-only heuristic (`composite_load_index` with thresholds). If the ML model does not beat the simple rules by a meaningful margin, we say so and ship the rules — that is the intellectually honest position and judges respect it.

9.4 Guardrails against the two named failure modes

The PS calls out false positives and false negatives explicitly.

Against false positives: abstention on sparse data · calibration · capacity-matched thresholds · personal baselines (so "always works nights" isn't flagged) · suppression while an intervention is open · flags expire.

Against false negatives: three parallel tracks so a signal missed by one may be caught by another · changepoint detection catching shifts before levels breach · engagement drop treated as signal · the app's always-available self-report path as a human override the model can't block.

10. Privacy and security architecture

10.1 Defence in depth

Layer	Control
Collection	Consent gate before ingest. Data without consent is never collected, not merely filtered. Data minimisation — we justify every field.
Identity	Pseudonymisation at ingest. Token vault in a separate schema with separate credentials, envelope-encrypted (KMS/HSM in production).
Storage	AES-256 at rest. Field-level encryption for names, service numbers, free-text self-reports, clinical assessments.
Transit	TLS 1.3. Certificate pinning in the mobile app.
Access	RBAC with field-level response filtering. Reason-for-access required. MFA for dashboard roles.
Aggregation	k-anonymity ($k \geq 10$) on every group view. l-diversity on sensitive attributes. Differential privacy noise on published unit statistics (P1).
Compute	Federated / on-device aggregation for biometrics (P2) — raw physiological streams never leave the handset.
Audit	Hash-chained append-only log of every individual access, with integrity verification endpoint.
Purpose	Technical block on export to promotion/posting/disciplinary/ACR systems.
Retention	Automated deletion schedule. Raw self-reports 12 months, derived features 24 months, audit logs 7 years.
Deployment	On-premise / air-gapped. No third-party AI APIs in the data path. Data residency: India.

10.2 Federated learning for the biometric track — P2, designed not built

Our most defensible architectural claim, and worth explaining well even though we won't implement it in the hackathon:

Wearable data is the most intimate data in the system. Rather than shipping HRV streams to a server, the model travels to the device: local training on the handset, only encrypted gradient updates transmitted, secure aggregation across ≥ 50 devices so no individual's update is recoverable, differential-privacy noise on the aggregate. The server never sees one person's heartbeat.

We build a **stub** that demonstrates the flow with mock updates, and document the full design. That is a legitimate P2 position.

10.3 DPDP Act 2023 mapping

Write this table into the presentation; judges will ask.

DPDP requirement	Our implementation
Notice before consent	Onboarding screens, plain Hindi/English, per-scope explanation
Free, specific, informed, unconditional, unambiguous consent	Four independent toggles, no dark patterns, app fully usable with all off
Right to withdraw consent	One tap in Privacy Centre; withdrawal is as easy as granting
Purpose limitation	PurposeGuard middleware with a hard block on career-affecting purposes
Data minimisation	Field-level justification register; disciplinary data explicitly excluded
Accuracy	Personnel can view and contest their own data; appeals mechanism
Storage limitation	Automated retention/deletion jobs, monitored in admin console
Reasonable security safeguards	Encryption at rest/transit, RBAC, audit, threat model

Breach notification	Documented incident-response procedure with timelines
Right to access	Personnel see their own data and their own access log
Right to erasure	"Delete my data" in Privacy Centre, cascading through all stores
Data Fiduciary obligations	DPO role defined; DPIA completed
Children's data	Not applicable — all data subjects are adult serving personnel

11. Anti-stigmatisation design standard

Named explicitly in the PS as a key technical challenge. It is a *design* deliverable, and these are its rules. Treat them as binding.

1. **Bands, never numbers, in any human-facing view.**
2. **Neutral vocabulary.** Say "check-in recommended." Never "at-risk," "high stress," "psychological concern," "flagged," "case." Vocabulary is enforced by a shared i18n string file — there is no other place to write UI copy.
3. **No rankings, no leaderboards, no sorting personnel by severity.** The queue is ordered by *urgency of action*, and this distinction is visible in the UI.
4. **Flags expire.** No permanent label. A cleared flag disappears from the person's record view.
5. **Aggregate-first defaults.** The dashboard opens on the unit, never on a list of people.
6. **Commander sees state, not reason.** "Welfare action in progress" and nothing more.
7. **Framing is supportive in every string.** The person's own view says "Your workload has been high recently — you're due for leave," never "You have been identified as stressed."
8. **Universal touchpoints.** Everyone gets check-in prompts, so opening the app is not itself a signal that you are struggling.

9. **Calm visual language.** No red alarms attached to a human being. Amber at most, and muted.
 10. **Interventions are normal, not exceptional.** Duty rotation and leave sanction are routine management actions, and the UI presents them as such.
-

12. Technology stack

Chosen for: one language family across the team, strong ML support, offline capability, and a deployment story that fits an air-gapped force network.

Layer	Choice	Why
Backend API	Python 3.11 + FastAPI	Same language as the ML stack — no cross-service serialization tax. Async, auto OpenAPI docs, Pydantic validation is exactly our data-contract need.
ML	scikit-learn, LightGBM , PyTorch (P1), SHAP, ruptures	LightGBM is the correct default for tabular; SHAP integrates natively; ruptures gives changepoints for free.
Experiment tracking	MLflow	Model versioning is a compliance requirement here, not a convenience.

Database	PostgreSQL 16 (+ TimescaleDB, P1)	Relational fits our schema; row-level security and column grants let us enforce RBAC at the DB layer too; append-only audit table via revoked grants.
Cache / queue	Redis	Alert deduplication, rate limiting, session store.
Job scheduling	APScheduler (demo) → Prefect (production)	Weekly feature/scoring jobs.
Dashboard	React 18 + TypeScript + Vite , TailwindCSS, shadcn/ui, TanStack Query, Recharts	Fast to build, good component ecosystem, type safety across the API boundary.
Mobile	React Native + Expo , TypeScript, expo-sqlite, i18next	Shares TS and mental model with the dashboard. Expo ships an Android build quickly. SQLite gives us offline-first.
Auth	JWT (short-lived) + refresh, TOTP MFA for dashboard roles	Self-contained, works air-gapped, no external IdP dependency.
Crypto	cryptography (Fernet/AES-GCM), envelope encryption, SHA-256 chain	Standard, auditable. HSM/KMS in production.
Containerisation	Docker + Docker Compose	The whole system stands up with one command — and that <i>is</i> the air-gapped deployment story.

Testing	pytest, Vitest, Playwright (E2E for the RBAC demo)	The RBAC block deserves an automated test we can show.
Docs	This file + OpenAPI + model card	—

Explicitly rejected: external LLM APIs anywhere in the data path (defeats air-gap and data residency); deep learning as the primary model (unjustifiable on tabular data of this size, and harms explainability); microservices sprawl (one modular monolith + one ML service is right for this scope); NoSQL (our data is deeply relational and we need referential integrity for audit).

13. API surface (the ones that matter)

— Personnel-facing —		
POST	/api/v1/checkins	submit daily
POST	/api/v1/assessments	submit PSS-10
GET	/api/v1/me/trends	own data back
GET	/api/v1/me/visibility	☆ "what my of
GET	/api/v1/me/access-log	who viewed my
POST	/api/v1/me/consent	grant / revok
POST	/api/v1/me/export	data export
DELETE	/api/v1/me/data	erasure requ
POST	/api/v1/support/contact	confidential
— Welfare officer —		
GET	/api/v1/units/{id}/overview	aggregate, k-
GET	/api/v1/queue	prioritised c
GET	/api/v1/personnel/{token}/detail	△ requires X-
POST	/api/v1/interventions	record decisi
PATCH	/api/v1/interventions/{id}/outcome	close the loo
— Commander —		
GET	/api/v1/units/{id}/workload	roster imbala

GET	/api/v1/units/{id}/status	counts only,
— Counsellor —		
GET	/api/v1/cases	assigned case
GET	/api/v1/cases/{id}/clinical	consent-gated
— Auditor —		
GET	/api/v1/audit/events	filtered acc
GET	/api/v1/audit/verify-chain	hash-chain in
— Admin —		
GET/PUT	/api/v1/admin/thresholds	band tuning p
GET	/api/v1/admin/models	model version
GET	/api/v1/admin/retention	deletion job
— Deliberately included so we can demo the block —		
POST	/api/v1/export/career-record	⬤ always retu

14. Demo script — the five scenarios

Build the prototype *around these*, in this order. Everything else is supporting material.

Scenario 1 — The gradual case (the core value proposition).

Constable CRP-A-0417. Over six weeks his night-duty share doubles against his own baseline, two leave applications are declined, and his weekly check-in scores drift down. The system moves him to **Check-in** recommended in week 6, listing three drivers. The officer sanctions home leave. Over the following weeks the band returns to **Stable**. **The line to say:** *"We flagged this 23 days before the point where the trajectory becomes a welfare incident. That is the whole product."*

Scenario 2 — The unit case (systemic, not individual). Following a critical incident, an entire section trends amber simultaneously. Instead

of emitting 40 individual referrals, the engine emits a **unit-level** recommendation: post-incident debrief and roster rebalancing. **The line:** *"When 40% of a company is amber, the problem is the duty roster, not forty individuals. A system that referred all forty to one counsellor would be useless."*

Scenario 3 — The crisis case (information barrier holds). A user reports acute distress in the app. It routes **immediately to the counsellor**, bypassing the chain of command. We then switch to the commander's dashboard and show it displaying only "welfare action in progress" — no reason, no detail. **The line:** *"If a jawan believes his CO will read his distress report, he will never file one. So his CO cannot read it."*

Scenario 4 — The privacy case (☆ highest-value 30 seconds of the demo). A welfare officer attempts to open an individual record without stating a reason — blocked with HTTP 428. They state a reason, access is granted, and the access appears instantly in the auditor's log. Then we run `verify-chain` and show the hash chain is intact. Finally we call `/export/career-record` and it returns 403 `PURPOSE_BLOCKED`. **The line:** *"Privacy here is not encryption. It is information architecture, and it is enforced in code."*

Scenario 5 — The consent case (revocation is real). A user revokes biometric consent in the app. On the dashboard, that person's assessment recomputes without biometric features, the confidence indicator drops to `REDUCED`, and the biometric drivers disappear from their card. Nothing is silently retained. **The line:** *"Revoking consent visibly changes what the system knows. That is the difference between a consent checkbox and a consent architecture."*

15. Repository structure

```
prahari/
```

```

└─ README.md
└─ CONCEPT.md ← this document
└─ docker-compose.yml
└─ docs/
    └─ model-card.md
    └─ dpia.md
    └─ dpdp-mapping.md
    └─ threat-model.md
    └─ escalation-sop.md
    └─ privacy-notice-en.md
    └─ privacy-notice-hi.md
└─ packages/
    └─ datagen/ # M1 - synthetic genera
        └─ configs/
        └─ causal_model.py
        └─ emitters/
        └─ validate.py
    └─ mock_hrms/ # M1 - mock source syst
    └─ ingest/ # M2 - adapters, ETL, p
        └─ adapters/base.py
        └─ adapters/mock.py
        └─ contracts/
        └─ pseudonymise.py
    └─ features/ # M3 - feature store
        └─ registry.py
        └─ leave.py duty.py deployment.py selfreport.
        └─ baselines.py changepoint.py
    └─ models/ # M4 - risk models
        └─ supervised.py anomaly.py fusion.py
        └─ calibration.py banding.py abstention.py
        └─ evaluate.py
    └─ explain/ # M5 - SHAP → sentences
        └─ driver_templates.yaml
    └─ recommend/ # M6 - intervention eng
        └─ rules.yaml
    └─ alerts/ # M7 - tiered router
    └─ consent/ # M8
    └─ governance/ # M9 - RBAC, audit chai
        └─ api/ # FastAPI app, routers,
└─ apps/

```

```

|   ├── dashboard/           # M10 – React + TS
|   └── mobile/              # M11 – React Native +
└── notebooks/              # EDA, model dev, eval
    └── tests/
        ├── test_rbac.py      # the demo-critical tes
        ├── test_purpose_guard.py
        ├── test_audit_chain.py
        └── test_k_anonymity.py

```

16. Risk register

Risk	Impact	Mitigation
Synthetic data is too easy → model looks fake-good	High	Validate generator (7.5); report baseline comparison; deliberately add noise/missingness
We build surveillance and call it welfare	Fatal	The five principles; anti-stigmatisation standard; Scenarios 3–5
Judges ask "where is your real data?"	High	Section 9.1 answer, rehearsed; public-dataset validation of physio components
Mobile app scope explodes	Medium	Only 5 screens are P0; self-help library is static content
Model becomes the whole project, privacy becomes a slide	High	Scenarios 4 and 5 are built <i>before</i> model tuning
Offline sync bugs eat two days	Medium	Simple last-write-wins by timestamp; queue-and-replay; no CRDTs
Demo depends on live model training	High	Pre-compute all assessments; demo reads from DB

Explainability produces useless drivers	Medium	Actionability filter; hand-written templates, not raw SHAP output
---	--------	---

17. Build plan

Ordered by **dependency**, not by team member. Everyone should understand every phase; work in pairs and rotate so knowledge stays shared.

Phase 0 — Foundation (Days 1–3)

- Repo, Docker Compose (Postgres + Redis + API skeleton), CI
- Schema migrations for Section 6 entities
- Read this document together; agree the five principles out loud
- **Gate:** `docker compose up` gives a running API and a migrated DB

Phase 1 — Data spine (Days 4–8)

- `datagen` causal model + emitters + config
- Generator validation (7.5)
- Mock HRMS service
- **Gate:** 2,000 personnel × 104 weeks generated; distributions sanity-checked; ground truth held out

Phase 2 — Ingest & governance core (Days 9–13)

- Adapter interface + mock adapter + Pydantic contracts + quarantine
- Consent Service (M8) — **build this before ingest consumes it**
- Pseudonymisation + token vault
- RBAC, hash-chained audit log, PurposeGuard (M9)
- **Gate:** `test_rbac.py`, `test_audit_chain.py`,

`test_purpose_guard.py` green. **Scenario 4 is now demoable.**

Building governance in Phase 2 rather than at the end is a deliberate choice. If it slips to the end it will not get built, and it is the part that wins this problem statement.

Phase 3 — Features (Days 14–18)

- Feature registry + all six feature families
- Personal baselines + z-scores + CUSUM changepoints
- Weekly materialisation job
- **Gate:** `features_weekly` populated for all personnel; feature EDA notebook committed

Phase 4 — Models (Days 19–25)

- Track A LightGBM with grouped time-series CV
- Track B Isolation Forest on baseline deviations
- Track C changepoint triggers
- Fusion, isotonic calibration, abstention, banding
- Full evaluation suite (9.3) incl. lead time, calibration curve, fairness audit
- Rules baseline for comparison
- **Gate:** evaluation report committed with lead-time number and calibration plot

Phase 5 — Explain, recommend, route (Days 26–30)

- SHAP → driver templates (EN/HI)
- Rules-based recommendation engine + capacity-aware prioritisation + unit-level rule
- Tiered alert router + fatigue controls + crisis pathway
- **Gate:** Scenario 2 demoable end-to-end

Phase 6 — Dashboard (Days 31–38)

- Unit Overview with k-anonymity → Check-in Queue → Individual View with reason-gate → Intervention Tracker → Workload → Counsellor Console → Audit View
- **Gate:** Scenarios 1, 2, 4 fully demoable through the UI

Phase 7 — Mobile (Days 39–45)

- Consent onboarding → Daily check-in → **"What my officer can see"** → My Trends → Talk to Someone → Privacy Centre
- Offline queue + sync
- Hindi localisation
- **Gate:** Scenarios 3 and 5 demoable across app + dashboard

Phase 8 — Governance docs & polish (Days 46–50)

- Model card, DPIA, DPDP mapping, threat model, privacy notices, escalation SOP
- Federated-learning stub + design doc (P2 story)
- Demo rehearsal ×5, timing each scenario
- **Gate:** full 8-minute run-through without touching a terminal

Standing rules

- Nobody merges to `main` without another person reading the diff — that is how knowledge stays shared without dividing labour
- Every phase ends with a 30-minute walkthrough where whoever did *not* build it explains it back
- If a phase slips, cut P1 scope, never cut governance

18. Open questions to settle as a team

1. Which force do we model concretely for the demo — a CRPF battalion in an LWE area is the most vivid choice
2. Do we demo in Hindi or English, and which do we screenshot for

the slides

3. Do we build the sequence autoencoder (P1) or stop at Isolation Forest
 4. What capacity numbers do we assume — one counsellor per battalion is defensible; confirm and cite
 5. Do we implement differential privacy on aggregates or document it as P2
 6. Who rehearses which demo scenario (everyone should be able to run all five)
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19. The one-paragraph pitch

Uniformed forces already collect the data that predicts occupational stress — leave patterns, duty rosters, deployment history, transfer records — but they read it manually, one commander at a time, far too late. Prahari reads that administrative exhaust against each individual's own baseline, detects the change before the crisis, and hands a welfare officer a short list of people to talk to, with the reason attached and a suggested action. It never shows a score against a name, never routes distress up the chain of command, never touches career decisions, and lets any person revoke any part of it in one tap. It flags earlier than a human can and forgets faster than a rumour does.

End of document. If something here is unclear, that is a bug in the document — raise it and we fix it.